

# MADISON M. BETTS

(724) 599-4122 | mmbetts03@gmail.com

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## EDUCATION

**M.Sc. in Fisheries and Wildlife Science**, *Virginia Tech, Blacksburg, VA* 07/2023

**B.Sc. in Environmental Science/Theatre Minor**, *Allegheny College, Meadville PA* 05/2021

## Experiential Learning

Caribbean Ecosystem Field Studies Program (ECOFS), Univ. of Montana, Xpu-Ha, Mexico. 2019

Coastal Marine Education and Research Academy (CMERA), Clearwater, FL. 2018

## WORK EXPERIENCE

**Graduate Research/Teaching Assistant** 01/2026 - Present

*University of Vermont, Burlington, VT*

*Supervisor: Dr. Thaïs Bernos | 20-40 hrs/wk*

**Fisheries Lab Geneticist** 11/2023 – 10/2025

*Lynker Technologies, contracted with NOAA NWFSC*

*Supervisor: Dr. Krista Nichols | 40 hrs/wk*

- Extracted DNA from a range of sources, including fresh and degraded tissues and eDNA filters
- Performed a variety of molecular lab techniques including PCR amplification, qPCR, library prep for GTseq and metabarcoding, and sequencing using Sanger and MiSeq machines
- Conducted field sampling and health monitoring for salmonid and anadromous fish studies
- Coordinated with NOAA trawl surveys and observers to collect specimens for the UW marine fish voucher collection; managed ID, tissue sampling, and specimen archiving with UW
- Generated genetic data using bioinformatics pipelines
- Supported the eDNA initiatives of the Integrated West Coast Pelagic Survey by conducting field work at sea, on board the Bell. M. Shimada

**Graduate Research/Teaching Assistant** 08/2021 – 08/2023

*Virginia Tech, Blacksburg, VA*

*Supervisor: Dr. Emmanuel Frimpong | 20-40 hrs/wk*

## As Lab Manager:

- Coordinated multiple field and lab-based biological studies to meet grant requirements
- Supervised and trained a 12-person field team during summers 2022 and 2023
- Contributed to annual funder reports and prepared manuscripts for publication
- Delegated work responsibilities to crew and managed work schedule
- Built outreach partnerships with local schools and organizations; delivered educational presentations

## As Research Assistant:

- Maintained long-term datasets and conducted daily habitat monitoring
- Designed and ran lab-based experiments on fish behavior and physiology
- Led genetic analysis for multiple projects; first in 5 years to sequence fish embryos
- Performed and trained lab technicians to complete DNA extractions, PCR amplification, and library preparation for sequencing

- Identified fish nests and spawning fishes to species using morphology, often from shore
- Led snorkel, seining, and electrofishing surveys in streams and rivers
- Contributed to annual funder reports; prepared manuscripts for publication

#### **As Teaching Assistant:**

- Taught labs in Ichthyology and Fish Ecology; led dissections (lamprey, shark, perch)
- Instructed students in species identification and field sampling (electrofishing, seining)
- Gave lectures on coral reef ecology, fish physiology and behavior
- Provided individual support during weekly office hours

#### **Fisheries Technician**

06/2021 - 08/2021

*Virginia Tech, Blacksburg, VA*

*Supervisor: Dr. Emmanuel Frimpong | 40 hrs/wk*

- Monitored larval development of multiple minnow species across six lab-based experiments
- Conducted electrofishing surveys in 1st–4th order streams throughout southwest Virginia and the Shenandoah Valley
- Performed stream habitat assessments and recorded daily abiotic data (flow, temperature, dissolved oxygen, pH)
- Measured oxygen consumption in larval fishes using Unisense micro-respirometry systems

#### **Avian Field Technician**

02/2020 - 04/2020

*IUP Research Institute, Indiana, PA*

*Supervisor: Dr. Jeff Larkin | 20 hrs/wk*

- Identified woodcock and other bird species using visual observation and spotting scopes
- Assisted in testing and refining transect survey protocols
- Recorded biological data accurately and consistently
- Conducted fieldwork in varied weather and rugged terrain, often involving moderate-distance hiking

#### **Environmental Science Dept. Office Assistant**

08/2017 - 05/2021

*Allegheny College, Meadville, PA*

*Supervisor: Ms. Ruth Dunton | 20 hrs/wk*

- Collaborated with Dept. faculty to schedule events and professional talks
- Developed skills to work independently and as part of a team
- Promoted department with various forms of social media
- Handled general office organization/paperwork

## **PUBLICATIONS**

Stahl, J., M. Hutchinson, J. Tucker, F. O'Neill, K. Parsons, **M. Betts**, C. Young, and E. Crigler. A comprehensive approach to collecting species-specific data and assessing post-release survival on Mobulid rays in the Hawai'i longline fisheries. In prep.

**Betts, M. M.**, E. A. Hultin, E. M. Hallerman, E. G. Maurakis, and E. A. Frimpong. 2026. Embryonic selfish-herding blurs the line between brood parasitism and mutualism for communal-breeding stream fishes. Accepted at *Ecology*, awaiting pre-print.

**Betts, M. M.**, D. C. Haak, E. M. Hallerman, E. G. Maurakis, and E. A. Frimpong. 2025. Mr. Mom: *Nocomis* embryo-burying effectively increases embryo survival amidst predation by all mutualism participants. *Ecology of Freshwater Fish*, 35(1). <http://dx.doi.org/10.1111/eff.70035>

Shelton, A. O., K. Nichols, K. M. Parsons, **M. M. Betts**, S. P. Engster, A. Ramón-Laca, M. Parsley, M. Shaffer, A. Wells. 2025. Developing an abundance index for pacific hake using environmental DNA (Appendix G in: Status of the Pacific Hake (whiting) stock in U.S.

Betts

and Canadian waters in 2025). 286 p.

**Betts, M. M.**, N. Abaid, E. G. Maurakis, and E. A. Frimpong. 2024. Bluehead chub *Nocomis leptocephalus* hosts exploit selfish-herd benefits from their heterospecific nest associates. *Freshwater Biology*, 69, 450–459. <https://doi.org/10.1111/fwb.14224>

**Betts, M. M.** 2023. Two Layers of Selfish-herds in Spawning Aggregations of Chub (*Nocomis sp.*) and its Nest Associates. Master's Thesis, Virginia Tech.

## GRANTS

**Betts M. M.**, et al. 2022. Bridging Stream and City: Inspiring Natural Wonder through Art and Technology (Outreach Grant). Institute for Creativity, Arts, and Technology (ICAT) Major SEAD Grant, Virginia Tech (\$23,000).

## AWARDS

Environmental Science Faculty Prize, Allegheny College, Meadville, PA, 2021

Distinguished Alden Scholar, Allegheny College, Meadville, PA, 2019-20

2x Alden Scholar, Allegheny College, Meadville, PA, 2017-18, 2018-19

## PRESENTATIONS & PUBLISHED ABSTRACTS

Yang, H., T. Bustamante, S. Kraus, S. Brooks, **M. Betts**, E. Frimpong, K. Strom, & H. Foroutan. *Combining Lab, Field, and Computational Approaches to Investigate the Transport of Dissolved Oxygen through Gravel-Mound Nests*. AGU Fall Meeting 2024, Washington, D.C., Poster No. 1414, id. EP51F-1414. 12/2024

*Two Layers of Selfish-Herds in the Spawning Aggregations of Chub and its Associates*. NOAA Conservation Biology Genetics & Evolution Program meeting, Seattle, WA (invited speaker). 02/2024

*A day in the life of one fascinating fish*. Kids Tech University at Virginia Tech, Blacksburg, VA (invited speaker). 02/2023

*Bridging Stream and City: Inspiring Natural Wonder through Art and Technology*. ICAT Playdate, Blacksburg, VA. 02/2023

*Selfish nesting behaviors of the Bluehead Chub, a keystone stream fish*. AFS 2022, Spokane, WA. 08/2022

Bustamante, T., **M. Betts**, S. Brooks, & E. Frimpong. *No Stone Unturned: Studying bluehead chubs at Virginia Tech*. New River Valley Symposium, New River Gorge National Park and Preserve, WV. 04/2022

*The Intersection Between Performance Art and Science*. VT Communicating Science Convention, Blacksburg, VA (invited panelist). 03/2022

*Discover Chub: Builders of Deerfield Trail*. VT Communicating Science Convention, Blacksburg, VA. 03/2022

*Predation response to chemosensory and visual cues in wild vs. cultured Chesapeake logperch* (*Percina bimaculata*). 13th Annual Erie Regional Science Consortium, Erie, PA. 11/2020

## OUTREACH

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|----------------------------------------------------------------------------------------------------------------------------|---------|
| Guest Lecture – Brevard College, Brevard, NC. (20 students)                                                                | 12/2025 |
| Skype a Scientist – Microbiology Girls Club, Seattle, WA. (20 students).                                                   | 07/2025 |
| NOAA Flavor of Fisheries Seminar 2024, Seattle, WA. (20 participants).                                                     | 08/2024 |
| Blacksburg Girl Scout Troup Meeting, Blacksburg, VA. (30 children and parents).                                            | 05/2023 |
| Girls Launch with Center for Communicating Science, Pembroke, VA. (75 students).                                           | 05/2023 |
| Montgomery County Stormwater Days, Christiansburg, VA. (378 students and chaperones).                                      | 04/2023 |
| Mountain Valley Pipeline Ecotheatre Workshop, Moss Arts Center, Blacksburg, VA. (15 participants, 65 audience)             | 11/2022 |
| Virginia Tech Science Fair, Moss Arts Center, Blacksburg, VA. (150 children and parents)                                   | 11/2022 |
| Pearisburg Public Library, Pearisburg, VA (60 children and parents).                                                       | 07/2022 |
| “Discover Chub: An Evening Experience,” Public outreach event, Blacksburg, VA. (70 community members)                      | 06/2022 |
| Art Installment, Student Arts Spotlight, Moss Arts Center, Blacksburg, VA. (200+ viewers).                                 | 12/2021 |
| “Touring the castles of hornyheads in a stream near you,” Master Naturalist Event, Blacksburg, VA. (35 community members). | 11/2021 |
| “Commit to Unwavering Curiosity,” VT Aspire! Awards, Blacksburg, VA. (75 VT faculty and staff).                            | 10/2021 |
| “The Art and Science of Stream Biodiversity,” Science on Tap of the NRV, Blacksburg, VA. (65 community members).           | 07/2021 |

## SKILLS & CERTIFICATIONS

### Computer Skills:

- R
- Geneious
- Python
- MATLAB
- Microsoft Office

### Certifications:

- CPR/First Aid + AED
- PADI Rescue Diver
- Ontario Pleasure Craft Operator Card (Boater’s License)
- Washington Driver’s License

## MEMBERSHIP

**President:** Art for Environmental Justice (2022-23)

**Vice President:** VT’s Chapter of the American Fisheries Society (2022-23)

**Member:** Climate Action, Sustainability, and Energy Committee at VT (2022-23)

**Member:** American Fisheries Society (National Chapter) (2022-)

**Member:** Communicating Science Club (2021-2023)

## WEBSITES

Professional website - <https://maddiebetts.com/>

Github - <https://github.com/madisonbetts>

ResearchGate - <https://www.researchgate.net/profile/Madison-Betts-4>

Google Scholar – <https://shorturl.at/afP9m>

## REFERENCES

**Dr. Thaïs Bernos**

*University of Vermont*

Thais.Bernos@uvm.edu

**Dr. Emmanuel Frimpong**

*Virginia Tech*

frimp@vt.edu

(540) 231-6880

**Dr. Meghan Parsley**

*University of Idaho*

mbparsley@uidaho.edu

(614)-545-8538